MATHEMATICS

Maximum Marks: 80 **Time Allowed: Three Hours**

Reading Time: Additional Fifteen Minutes

Instructions to Candidates

1. You are allowed an additional **fifteen minutes** for **only** reading the paper.
2. You must **NOT** start writing during reading time.
3. The question paper has **11** printed pages.
4. It consists of **twenty** questions and **four** sections: Sections A, B, C and D.
5. All questions are compulsory.
6. Section A consists of very short answer questions of 1 mark each.
7. Section B consists of short answer questions of 2 marks each.
8. Section C consists of moderately long answer questions of 3 marks each.
9. Section D consists of long answer questions of 5 marks each.
10. Internal choices have been provided in **three** questions, each in Sections B, C and D.
11. While attempting Multiple Choice Questions in Section A, you are required to write only **ONE** option as the answer.
12. The intended marks for questions or parts of questions are given in the brackets [ ].
13. All workings, including rough work, should be done on the same page as, and adjacent to, the rest of the answer.
14. Mathematical tables and graph papers are provided.

Instruction to Supervising Examiner

Kindly read aloud the instructions given above to all the candidates present in the examination hall.

SECTION A – 20 MARKS

Question 1

In subparts (i) to (xvii) choose the correct options and in subparts (xviii) to (xx), answer the questions as instructed.

(i) The relation R on the set of integers defined by a R b if and only if $a^2 - 5ab + 4b^2 = 0$ is: [1]
(Understand)

(a) an equivalence relation

(b) symmetric and transitive but not reflexive

(c) reflexive but neither symmetric nor transitive

(d) reflexive and symmetric but not transitive

(ii) The principal value of $\cos^{-1}(\sin \frac{10\pi}{7})$ is: *(Apply)* [1]

(a) $-\frac{13\pi}{14}$

(b) $\frac{3\pi}{14}$

(c) $\frac{11\pi}{14}$

(d) $\frac{13\pi}{14}$

(iii) If A is a square matrix of order 3 and $|A| = -2$, then the value of $|-3A|$ is: *(Apply)* [1]

(a) 54

(b) −54

(c) 18

(d) −18

(iv) A cylindrical popcorn tub of radius 10 cm is being filled at the rate of 200π cm^3 per minute. The level of the popcorn in the tub is rising at the rate of: *(Apply)* [1]

(a) 0·5 cm/min

(b) 2 cm/min

(c) 1 cm/min

(d) 20 cm/min

(v) **Statement I:** $f(x) = 7 + x - e^x$ is strictly increasing in $(-\infty, 0)$. [1]

Statement II: $f(x) = 7 + x - e^x$ is strictly increasing in $(0, \infty)$.

(Understand)

(a) Statement I is true and Statement II is false.

(b) Statement I is false and Statement II is true.

(c) Both the statements are true.

(d) Both the statements are false.

(vi) The curves $y^2 = 4x$ and $xy = k$ cut each other at right angles. The value of k is: *(Apply)* [1]

(a) $\pm 2\sqrt{2}$

(b) ±4

(c) ±8

(d) $\pm 4\sqrt{2}$

(vii) For $f(x) = x^3 - 6x^2 + 9x + 15$, the local maximum value and the point where it occurs are: *(Apply)* [1]

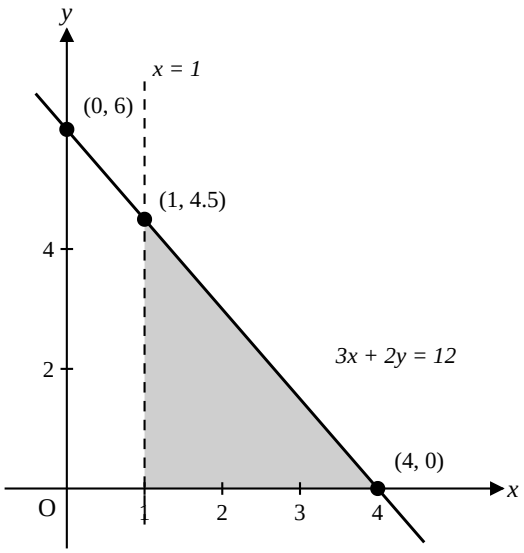
(a) 15 at x = 3

(b) 19 at x = 3

(c) 19 at x = 1

(d) 15 at x = 1

(viii) The area of the shaded region bounded by the line $3x + 2y = 12$, the line $x = 1$ and the x-axis is given by: *(Apply)* [1]



(a) $\int_1^4 (6 + \frac{3x}{2})\, dx$

(b) $\int_1^4 (6 - \frac{3x}{2})\, dx$

(c) $\int_0^6 (4 - \frac{2y}{3})\, dy$

(d) $\int_0^4 (6 - \frac{3x}{2})\, dx$

(ix) A student evaluates $\int_0^{2\pi} \sqrt{1 - \cos 2x}\, dx$ as follows. [1]

Step 1: $1 - \cos 2x = 2 \sin^2 x$, so the integrand is $\sqrt{2} \cdot \sqrt{\sin^2 x}$.

Step 2: $\sqrt{\sin^2 x} = \sin x$, so the integrand is $\sqrt{2} \sin x$.

Step 3: the integral $= [-\sqrt{2} \cos x]_0^{2\pi} = 0$.

In which step is there an error (if any)? *(Understand)*

(a) Step 1

(b) Step 2

(c) Step 3

(d) There is no error

(x) Two students evaluate $\int \sin x \cos x\, dx$. Riya writes $\frac{1}{2} \sin^2 x + C$ and Sameer writes $-\frac{1}{2} \cos^2 x + C$. Whose answer is correct? *(Analyse)* [1]

(a) Only Riya's

(b) Only Sameer's

(c) Both are correct

(d) Neither is correct

(xi) If $\vec{a} = 2\hat{i} + \hat{j} - 3\hat{k}$, $\vec{b} = \hat{i} - 2\hat{j} + 4\hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j} + 2\hat{k}$, then $(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c})$ is equal to: *(Recall)* [1]

(a) 138

(b) 128

(c) −138

(d) 148

(xii) The angle between the planes $x - y + 2z = 7$ and $2x + y + z = 4$ is: *(Apply)* [1]

(a) $\pi/6$

(b) $\pi/4$

(c) $\pi/2$

(d) $\pi/3$

(xiii) $Z = ax + by$, where $a, b \geq 0$ and $a + b = 8$, has its maximum value at both the corner points (6, 9) and (9, 4) of the feasible region. The maximum value of Z is: *(Apply)* [1]

(a) 45

(b) 57

(c) 72

(d) 60

(xiv) **Statement I:** If the feasible region of a linear programming problem is bounded, the objective function has both a maximum and a minimum value on it. [1]

Statement II: If an objective function has its maximum at two corner points, it has the same maximum at every point of the line segment joining them.

(Understand)

(a) Statement I is true and Statement II is false.

(b) Statement I is false and Statement II is true.

(c) Both the statements are true.

(d) Both the statements are false.

(xv) If $P(A \mid B) = 0{\cdot}4$ and $P(A \cap B) = 0{\cdot}14$, then $P(A' \cap B)$ is: *(Understand)* [1]

(a) 0·56

(b) 0·35

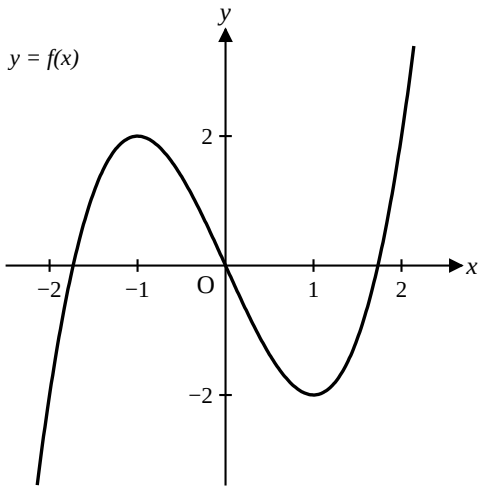
(c) 0·14

(d) 0·21

(xvi) **Assertion:** The function f : $\mathbb{R} \to \mathbb{R}$, $f(x) = x^3 - 3x$, whose graph is shown, is not one-one. [1]

Reason: If every line parallel to the y-axis cuts a curve at no more than one point, then the curve represents a function.

(Analyse)



(a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.

(b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.

(c) Assertion is true and Reason is false.

(d) Assertion is false and Reason is true.

(xvii) **Assertion:** $\int_0^{\frac{\pi}{2}} \sin x / (\sin x + \cos x)\, dx = \frac{\pi}{4}$. [1]

Reason: $\int_0^a f(x)\, dx = \int_0^a f(a - x)\, dx$ for a continuous function f.

(Understand)

(a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.

(b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.

(c) Assertion is true and Reason is false.

(d) Assertion is false and Reason is true.

(xviii) Find the value of x for which the matrix $A = \begin{bmatrix} 0 & 2 & -5 \\ -2 & 0 & 3 \\ x & -3 & 0 \end{bmatrix}$ is skew-symmetric. *(Understand)* [1]

(xix) The solution of the differential equation $\frac{dy}{dx} = -\frac{ax + 1}{3y - 2}$ represents a circle. Find a. *(Analyse)* [1]

(xx) The random variable X takes the values 1, 2, 3, 4, 5 with P(X = 1) = m, P(X = 2) = 2m, P(X = 3) = a, P(X = 4) = 3m and P(X = 5) = b. If P(X ≤ 2) = 0·15 and P(X ≥ 4) = 0·60, find P(X = 3). *(Apply)* [1]

SECTION B – 14 MARKS

Question 2

Find the angle between the line $\frac{x+1}{2} = \frac{y}{3} = \frac{z-3}{6}$ and the plane $10x + 2y - 11z = 3$. [2]

(Apply)

Question 3

A die has the faces 1, 2, 3 coloured red and the faces 4, 5, 6 coloured green. It is thrown once. Let A be the event "the number is a multiple of 3" and B the event "the colour is green". Show that A and B are independent events. *(Evaluate)* [2]

Question 4

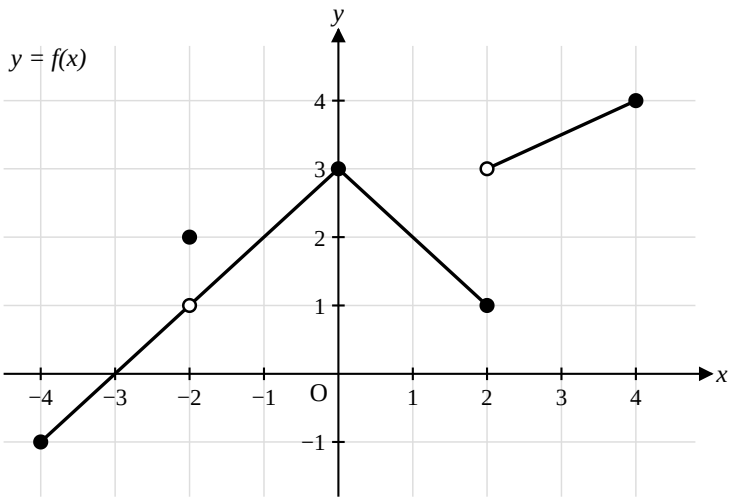
Solve the differential equation $x\,dy - y\,dx = \sqrt{x^2 + y^2}\,dx$, $x > 0$. *(Apply)* [2]

Question 5

The function $f(x) = x^2 + 3x + p$ for $x \le 1$ and $f(x) = qx + 2$ for $x > 1$ is differentiable at $x = 1$. Find p and q. *(Understand)* [2]

OR

The graph of a function f on $[-4, 4]$ is shown. (An open circle is a point that is not on the graph; a filled circle is.)



(a) At which point does f have a removable discontinuity? *(Understand)* [1]

(b) At which point is f continuous but not differentiable? *(Understand)* [1]

Question 6

A music app scores listening time with $f(x) = \tan^{-1}\frac{x}{12}$, where x is the hours listened per month. Two users listen for 4 hours and 9 hours. Find $f(4) + f(9)$. *(Apply)* [2]

OR

Solve for x: $\tan^{-1}x - \cot^{-1}x = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$. *(Apply)* [2]

Question 7

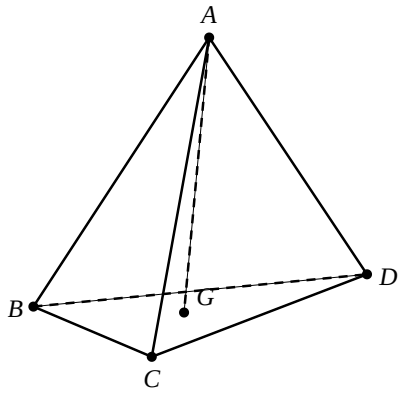
Find the unit vector perpendicular to both $\vec{a} = 3\hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{b} = 3\hat{i} + 4\hat{j} + \hat{k}$. *(Apply)* [2]

OR

Three drone cameras are at A(1, 5, 3), B(3, 4, 2) and C(7, 2, 0), with position vectors relative to the control tower O. Using vectors, show that the drones are on a straight path (A, B, C are collinear), and find the ratio in which B divides AC. *(Apply)* [2]

Question 8

A building is to be constructed in the form of a triangular pyramid ABCD (figure, not to scale). In metres, A(0, 4, 4), B(0, 2, 0), C(1, 5, 4) and D(5, 2, 2); G is the point of intersection of the medians of ΔBCD.



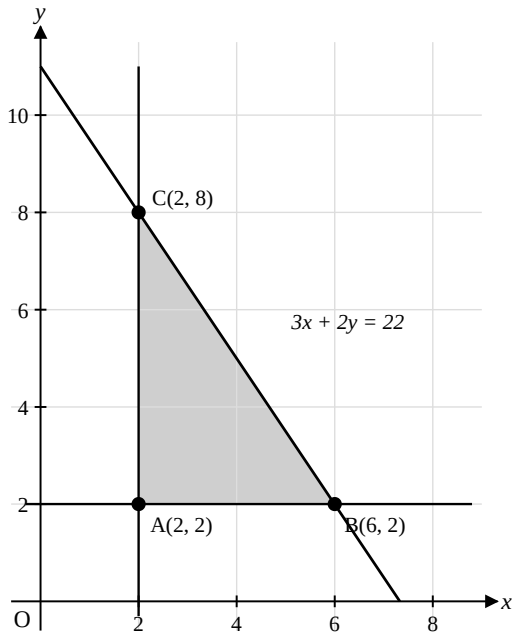
(i) Find the length of the vector $\vec{AG}$. *(Apply)* [1]

(ii) Find the area of ΔABC. *(Apply)* [1]

SECTION C – 21 MARKS

Question 9

The feasible region of a linear programming problem is the shaded triangle ABC shown in the figure.



(i) Write the constraints that define the feasible region. *(Understand)* [1]

(ii) Find the maximum value of Z = 2x + 3y and the point where it occurs. *(Apply)* [1]

(iii) If the constraint $3x + 2y \le 22$ is removed, does Z have a maximum value on the new region? Give a reason. *(Analyse)* [1]

Question 10

If x = sin t and y = sin 2t, prove that $(1 - x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + 4y = 0$. *(Apply)* [3]

Question 11

Let $f(x) = \tan^{-1}\frac{\sqrt{1 + x^2} - 1}{x}$, $x \neq 0$.

(i) Find $f'(x)$. *(Apply)* [2]

(ii) A student claims that f has a local maximum at some point c. Use f' to evaluate the claim. *(Evaluate)* [1]

Question 12

Two planes are π_1: $x + 2y - 2z = 3$ and π_2: $2x + 4y + pz = 12$. P is the point (1, 2, 1).

(i) Verify that P lies on π_1. *(Understand)* [1]

(ii) Find p if π_2 is parallel to π_1. *(Apply)* [1]

(iii) The line through P perpendicular to π_1 meets π_2 (with this p) at Q. Find Q. *(Analyse)* [1]

OR

Find the equation of the plane containing the line $\frac{x+1}{3} = \frac{y-2}{-1} = \frac{3-z}{2}$ and the point (2, 0, 1). *(Apply)* [3]

Question 13

Using properties of determinants, prove that $\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4abc$. *(Analyse)* [3]

OR

A school needs decorations for 120 chairs (one bow each) and 25 tables (one cover each). Company A charges ₹14 per bow and ₹160 per cover; Company B charges ₹12 per bow and ₹180 per cover. Using matrix multiplication, find the total cost for each company and state which is cheaper. *(Analyse)* [3]

Question 14

f is a continuous function on [2, 7] with $\int_2^7 4f(x)\,dx = 20$ and $\int_2^4 f(x)\,dx = 2$.

(i) Show that $\int_7^2 f(x)\,dx = -5$. *(Understand)* [1]

(ii) Find $\int_4^7 (x + f(x))\,dx$. *(Apply)* [2]

Question 15

Using integration, find the area of the region enclosed by the parabola $y = 4 - x^2$ and the line $y = -2x + 4$. *(Apply)* [3]

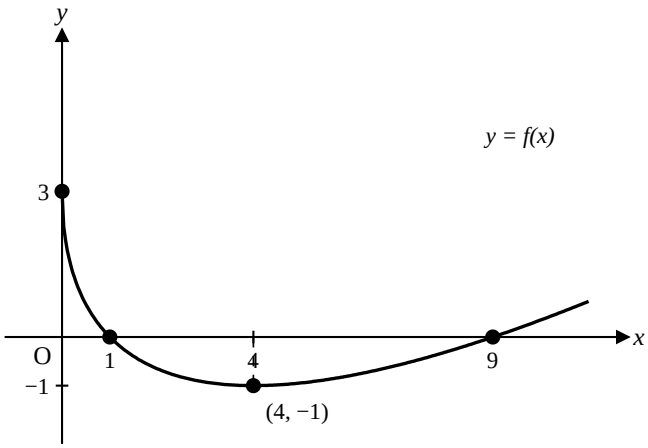
OR

Find the area bounded by the curve $y = x(3 - x)$ and the x-axis from $x = 0$ to $x = 4$. *(Apply)* [3]

SECTION D – 25 MARKS

Question 16

The net daily gain of a bakery, in thousands of rupees, when it bakes x hundred loaves is modelled by $f(x) = x - 4\sqrt{x} + 3$. The graph of f is shown.



(i) Write the natural domain of f. *(Recall)* [1]

(ii) Does f have an inverse on its natural domain? Justify. *(Understand)* [1]

(iii) $g : [0, 4] \to [-1, 3]$ is defined by $g(x) = f(x)$. Find $g^{-1}(x)$ and verify that $g(g^{-1}(0)) = 0$. *(Apply)* [1]

(iv) $h : [4, \infty) \to [-1, \infty)$ is defined by $h(x) = f(x)$. Find $h^{-1}(x)$ and verify that $h(h^{-1}(3)) = 3$. *(Apply)* [1]

(v) Find the value of x for which $g^{-1}(x) = h^{-1}(x)$. *(Analyse)* [1]

Question 17

A closed cylindrical water tank holds 2156 cubic units of water when full. Its base radius is r and its height is h. Take $\pi = \frac{22}{7}$.

(a) Express h in terms of r. *(Understand)* [1]

(b) Express the total surface area S in terms of r. *(Apply)* [1]

(c) Prove that S is least when r = 7. *(Analyse)* [2]

(d) Find the height of the tank for the least surface area. *(Apply)* [1]

OR

A cable-car track rises and falls so that its height h(x), in hundreds of metres, at a horizontal distance x km from the start is $h(x) = x^3 - 9x^2 + 24x + 4$, $x \geq 0$.

(a) Find the height of the track at the starting point. *(Understand)* [1]

(b) Find the locations of the two turning points of the track and state which is a peak. *(Apply)* [2]

(c) Find the difference in height between the peak and the trough. *(Apply)* [1]

(d) Find the absolute maximum and absolute minimum heights of the track on [0, 3]. *(Evaluate)* [1]

Question 18

Answer the following.

(a) Evaluate $\int \log x \, dx$. *(Understand)* [2]

(b) Hence evaluate $\int \frac{\log(\tan^{-1}x)}{1+x^2}\,dx$. *(Evaluate)* [3]

OR

(a) Show that $\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$. *(Understand)* [2]

(b) Hence evaluate $\int \frac{\cos x}{\sin^2 x + 4\sin x + 8}\,dx$. *(Evaluate)* [3]

Question 19

A school offers three modes of attending classes: Mode A (offline) is used by 50% of the students, Mode B (live online) by 30% and Mode C (recorded lectures) by 20%. A survey finds that 40% of the students of Mode A, 50% of Mode B and 70% of Mode C rated their classes "satisfactory". A student is selected at random.

(a) Define suitable events and find the probability that the student rated the classes satisfactory. *(Apply)* [2]

(b) The student rated satisfactory. Find the probability that the student used Mode C. *(Analyse)* [2]

(c) Given that the student is satisfied, which mode is the most likely, and what is its probability? *(Evaluate)* [1]

OR

The English, Mathematics and Commerce teachers of a school give surprise tests on a given day independently, with probabilities 0·8, 0·6 and 0·5 respectively. Let X be the number of surprise tests a student gets on a day. The school wants the average number of tests per day to be at most 2; otherwise the teachers must coordinate.

(a) Write the possible values of X. *(Understand)* [1]

(b) Find the probability distribution of X. *(Apply)* [2]

(c) Find the average number of tests per day. *(Apply)* [1]

(d) Should the teachers coordinate? *(Evaluate)* [1]

Question 20

A furniture factory makes tables, chairs and cots from three kinds of wood. The tonnes of wood needed for one piece of each are: [5]

	Table	**Chair**	**Cot**
Teakwood	2	3	4
Rosewood	1	1	2
Satinwood	3	2	1

The factory has 28 tonnes of teakwood, 13 tonnes of rosewood and 17 tonnes of satinwood, and uses all of it. If x tables, y chairs and z cots are made, translate the data into a matrix equation, solve it by the matrix method (using A^{-1}) and hence find how many of each are made. *(Evaluate)*

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